

AP Statistics MCQ Practice Exam 01

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AP Statistics MCQ Practice Exam 1

PART A, QUESTIONS

APSTAT-MCQ1-001

Section: MCQ

Type: mcq

Unit: Unit 1, Exploring One-Variable Data and Collecting Data

Skill: Analyze Data

Prompt:

A battery laboratory recorded the number of full charge cycles completed by seven prototype cells before capacity fell below a required level:

410, 420, 425, 430, 435, 440, 515

Using the median-of-halves method, which statement correctly identifies potential outliers by the $1.5 \times$ IQR rule?

- A) There are no potential outliers because 515 is less than 3 standard deviations above the mean.
- B) The value 515 is a potential outlier because it is greater than the upper fence of 470 cycles.
- C) The value 515 is a potential outlier because it is greater than the upper fence of 460 cycles.
- D) Both 410 and 515 are potential outliers because they are the minimum and maximum.

APSTAT-MCQ1-002

Section: MCQ

Type: mcq

Unit: Unit 3, Inference for Categorical Data: Proportions

Skill: Interpret Results

Prompt:

An independent random sample of subscribers from each of two streaming plans was used to estimate the difference $p_P - p_S$, where p_P is the proportion of all premium-plan subscribers who renew and p_S is the corresponding proportion for the standard plan. A 95% confidence interval for $p_P - p_S$ is (0.04, 0.18). Which interpretation is best supported?

- A) Ninety-five percent of premium subscribers renew at a rate between 4% and 18% higher than standard subscribers.
- B) The data provide evidence that the premium-plan population renewal proportion is between 4 and 18 percentage points higher than the standard-plan proportion.

- C) In 95% of months, the premium renewal proportion exceeds the standard renewal proportion by 0.04 to 0.18.
- D) There is a 95% probability that the sample difference in renewal proportions is between 0.04 and 0.18.

APSTAT-MCQ1-003

Section: MCQ

Type: mcq

Unit: Unit 2, Probability, Random Variables, and Probability Distributions

Skill: Analyze Data

Prompt:

At a parcel hub, 160 parcels were checked for a torn label and for a late scan. The results were:

	Late scan	Not late	Total
Torn label	18	12	30
Label not torn	22	108	130
Total	40	120	160

If one of the parcels with a torn label is selected at random, what is the probability that it also had a late scan?

- A) 0.113
- B) 0.450
- C) 0.188
- D) 0.600

APSTAT-MCQ1-004

Section: MCQ

Type: mcq

Unit: Unit 1, Exploring One-Variable Data and Collecting Data

Skill: Collect Data

Prompt:

A ferry agency wants to estimate the mean satisfaction rating of weekday riders. Morning riders make up 70% of weekday ridership, and evening riders make up 30%. Satisfaction may differ by time of day. Which method is most appropriate for producing a sample that represents both groups and permits an overall estimate?

- A) Ask the first 200 riders who enter the busiest morning ferry to complete the survey.
- B) Randomly select two weekday ferry trips and survey every rider on those trips.
- C) Take separate simple random samples of morning and evening riders, with sample sizes proportional to their shares of weekday ridership.

D) Post the survey link on the agency website and weight the responses by time of day.

APSTAT-MCQ1-005

Section: MCQ

Type: mcq

Unit: Unit 4, Inference for Quantitative Data: Means

Skill: Analyze Data

Prompt:

A museum claims that visitors spend a mean of 17 minutes in a new gallery. For a random sample of 25 visitors, $\bar{x}$ = 18.6 minutes and s = 3.5 minutes. For a one-sample t-test of $H_0: \mu = 17$ against $H_a: \mu > 17$, what is the test statistic?

- A) $t = 2.29$
- B) $t = 0.46$
- C) $t = 4.57$
- D) $t = 11.43$

APSTAT-MCQ1-006

Section: MCQ

Type: mcq

Unit: Unit 5, Regression Analysis

Skill: Analyze Data

Prompt:

A least-squares regression model predicts the number of minutes required to unload a delivery truck from the number of pallets. For a truck carrying 26 pallets, the model predicts 41.2 minutes. The observed unloading time was 37.0 minutes. What is the residual for this truck?

- A) -4.2 minutes
- B) 4.2 minutes
- C) 37.0 minutes
- D) 78.2 minutes

APSTAT-MCQ1-007

Section: MCQ

Type: mcq

Unit: Unit 3, Inference for Categorical Data: Proportions

Skill: Formulate Questions

Prompt:

A library director will introduce a self-checkout interface if there is convincing evidence that more than 70% of all checkout attempts can be completed without staff help. Let p be the population proportion of

attempts completed without help using the interface. Which hypotheses address the director's question?

- A) $H_0: p = 0.70$; $H_a: p \neq 0.70$
- B) $H_0: p > 0.70$; $H_a: p = 0.70$
- C) $H_0: p = 0.70$; $H_a: p < 0.70$
- D) $H_0: p = 0.70$; $H_a: p > 0.70$

APSTAT-MCQ1-008

Section: MCQ

Type: mcq

Unit: Unit 1, Exploring One-Variable Data and Collecting Data

Skill: Collect Data

Prompt:

A housing office has a list of 120 apartment buildings and wants to estimate the proportion of apartments with working smoke detectors. It randomly selects 8 buildings and inspects every apartment in each selected building. Which sampling method was used?

- A) A simple random sample of apartments
- B) A stratified random sample by building
- C) A cluster sample using buildings as clusters
- D) A systematic sample ordered by building

APSTAT-MCQ1-009

Section: MCQ

Type: mcq

Unit: Unit 2, Probability, Random Variables, and Probability Distributions

Skill: Analyze Data

Prompt:

Questions APSTAT-MCQ1-009 through APSTAT-MCQ1-011 use the following stimulus. Let X be the number of pump failures during a randomly selected week at an automated greenhouse. The probability distribution is:

x	0	1	2	3
$P(X = x)$	0.50	0.30	0.15	0.05

What is the mean of X ?

- A) 0.55 failure
- B) 0.75 failure
- C) 1.00 failure
- D) 1.50 failures

APSTAT-MCQ1-010

Section: MCQ

Type: mcq

Unit: Unit 2, Probability, Random Variables, and Probability Distributions

Skill: Analyze Data

Prompt:

Questions APSTAT-MCQ1-009 through APSTAT-MCQ1-011 use the following stimulus. Let X be the number of pump failures during a randomly selected week at an automated greenhouse. The probability distribution is:

x	0	1	2	3
$P(X = x)$	0.50	0.30	0.15	0.05

What is the standard deviation of X ?

- A) 0.56 failure
- B) 0.75 failure
- C) 0.79 failure
- D) 0.89 failure

APSTAT-MCQ1-011

Section: MCQ

Type: mcq

Unit: Unit 2, Probability, Random Variables, and Probability Distributions

Skill: Interpret Results

Prompt:

Questions APSTAT-MCQ1-009 through APSTAT-MCQ1-011 use the following stimulus. Let X be the number of pump failures during a randomly selected week at an automated greenhouse. The probability distribution is:

x	0	1	2	3
$P(X = x)$	0.50	0.30	0.15	0.05

Which statement correctly interprets the mean of X ?

- A) Over many weeks, the average number of pump failures per week will approach 0.75.
- B) During 75% of weeks, exactly one pump will fail.
- C) The greenhouse will have a pump failure after exactly 0.75 week.
- D) In any four-week period, exactly three pump failures will occur.

APSTAT-MCQ1-012

Section: MCQ

Type: mcq

Unit: Unit 4, Inference for Quantitative Data: Means

Skill: Interpret Results

Prompt:

Twelve graduate students recorded their weekly commuting time before and after moving to a new campus shuttle schedule. For each student, the difference was defined as after – before. A 95% confidence interval for the population mean difference is $(-1.8, -0.4)$ hours. Which conclusion is appropriate?

- A) Ninety-five percent of students reduced their commuting time by between 0.4 and 1.8 hours.
- B) There is no evidence of a change because zero was not used as an endpoint.
- C) The data indicate that the population mean weekly commuting time after the change is 0.4 to 1.8 hours lower than before.
- D) The mean commute of every student after the change is at least 0.4 hour lower than before.

APSTAT-MCQ1-013

Section: MCQ

Type: mcq

Unit: Unit 1, Exploring One-Variable Data and Collecting Data

Skill: Analyze Data

Prompt:

A sensor reports lake-water temperatures in degrees Celsius. Due to a calibration error, 12 must be added to every recorded value. What happens to the sample mean and sample standard deviation after the correction?

- A) The mean increases by 12, and the standard deviation is unchanged.
- B) The mean is unchanged, and the standard deviation increases by 12.
- C) Both the mean and standard deviation increase by 12.
- D) The mean is multiplied by 12, and the standard deviation is unchanged.

APSTAT-MCQ1-014

Section: MCQ

Type: mcq

Unit: Unit 3, Inference for Categorical Data: Proportions

Skill: Analyze Data

Prompt:

In a simple random sample of 200 independent restaurant takeout orders, 86 included a reusable utensil option. Which is the approximate 95% confidence interval for the population proportion of takeout orders that include the option?

- A) $(0.396, 0.464)$

- B) (0.361, 0.499)
- C) (0.353, 0.507)
- D) (0.430, 0.499)

APSTAT-MCQ1-015

Section: MCQ

Type: mcq

Unit: Unit 5, Regression Analysis

Skill: Interpret Results

Prompt:

For a sample of manufacturing machines, a linear regression used weekly operating hours to predict weekly repair cost. The coefficient of determination was $r^2 = 0.64$. Which statement correctly interprets this value?

- A) Sixty-four percent of machines had repair costs predicted exactly by the model.
- B) The correlation between operating hours and repair cost was 0.64.
- C) Increasing operating hours causes 64% of the increase in repair cost.
- D) Sixty-four percent of the variation in weekly repair cost is explained by its linear relationship with weekly operating hours.

APSTAT-MCQ1-016

Section: MCQ

Type: mcq

Unit: Unit 1, Exploring One-Variable Data and Collecting Data

Skill: Collect Data

Prompt:

An agronomist compares three soil treatments on a hillside where growing conditions change from the top to the bottom. There are 18 similar plots in each of three horizontal slope bands. Which design best controls for the location on the slope?

- A) Apply one treatment to every plot in the top band, a second to the middle band, and the third to the bottom band.
- B) Let the farm manager choose a treatment for each plot based on soil appearance.
- C) Within each slope band, randomly assign equal numbers of plots to the three treatments.
- D) Randomly select one slope band and assign all three treatments only within that band.

APSTAT-MCQ1-017

Section: MCQ

Type: mcq

Unit: Unit 2, Probability, Random Variables, and Probability Distributions

Skill: Analyze Data

Prompt:

In a large population of online orders, 30% are delivered on the day promised. For random samples of 400 orders, what are the mean and standard deviation of the sampling distribution of $\hat{p}$, the sample proportion delivered on the promised day?

- A) Mean 0.30 and standard deviation 0.000525
- B) Mean 0.30 and standard deviation 0.0229
- C) Mean 120 and standard deviation 9.17
- D) Mean 0.70 and standard deviation 0.0229

APSTAT-MCQ1-018

Section: MCQ

Type: mcq

Unit: Unit 4, Inference for Quantitative Data: Means

Skill: Analyze Data

Prompt:

Two independent random samples are used to estimate $\mu_A - \mu_B$. For sample A, $n_A = 36$ and $s_A = 4.8$. For sample B, $n_B = 49$ and $s_B = 5.6$. What is the estimated standard error of $\bar{x}_A - \bar{x}_B$?

- A) 1.13
- B) 1.49
- C) 2.08
- D) 7.36

APSTAT-MCQ1-019

Section: MCQ

Type: mcq

Unit: Unit 3, Inference for Categorical Data: Proportions

Skill: Collect Data

Prompt:

Twenty-four clinics volunteer for a study of a text-reminder system. The clinics are randomly assigned either to use the system or to continue their usual procedures. The response is whether each scheduled appointment is missed. Which scope of inference is most defensible if the reminder group has a significantly lower missed-appointment proportion?

- A) The system caused a reduction for all clinics because random assignment makes the volunteers representative.
- B) The system is associated with fewer missed appointments, but no causal conclusion is possible because clinics volunteered.
- C) The result can be generalized to all clinics, but causation cannot be concluded.

D) A causal effect can be concluded for the participating clinics, but generalization to all clinics is not justified.

APSTAT-MCQ1-020

Section: MCQ

Type: mcq

Unit: Unit 1, Exploring One-Variable Data and Collecting Data

Skill: Interpret Results

Prompt:

Side-by-side boxplots summarize delivery distances, in kilometers, for two bicycle couriers. Both have median 6.5 km. Courier A has $Q1 = 3$ km and $Q3 = 11$ km. Courier B has $Q1 = 5$ km and $Q3 = 8$ km and also has one high outlier at 18 km. Which comparison is correct?

- A) Courier B has the larger IQR because of the high outlier.
- B) Courier A has a larger median and a smaller IQR.
- C) Their medians and IQRs are equal, but only Courier B has a large range.
- D) Their medians are equal, and the middle half of Courier A's distances is more variable.

APSTAT-MCQ1-021

Section: MCQ

Type: mcq

Unit: Unit 5, Regression Analysis

Skill: Analyze Data

Prompt:

For a dataset relating x = insulation thickness in centimeters to y = heat loss in watts, $r = -0.72$, $s_x = 2.4$ cm, and $s_y = 9.0$ watts. What is the slope of the least-squares regression line for predicting heat loss from insulation thickness?

- A) -2.70 watts per centimeter
- B) -1.73 watts per centimeter
- C) 1.73 watts per centimeter
- D) 2.70 watts per centimeter

APSTAT-MCQ1-022

Section: MCQ

Type: mcq

Unit: Unit 2, Probability, Random Variables, and Probability Distributions

Skill: Formulate Questions

Prompt:

At a shop, each of 8 independently selected customers has probability 0.15 of returning an item within 30 days. Let X be the number of these customers who return an item. Which probability represents the question, "What is the probability that at least 3 of the 8 customers return an item?"

- A) $P(X = 3)$
- B) $P(X < 3)$
- C) $P(X \geq 3)$
- D) $P(X \leq 3)$

APSTAT-MCQ1-023

Section: MCQ

Type: mcq

Unit: Unit 3, Inference for Categorical Data: Proportions

Skill: Analyze Data

Prompt:

A researcher compares the proportion who support a transit proposal in two independent random samples. In district 1, 78 of 120 respondents support it; in district 2, 60 of 120 support it. For a test of $H_0: p_1 = p_2$, what is the approximate two-proportion z statistic?

- A) 1.66
- B) 2.35
- C) 2.65
- D) 3.33

APSTAT-MCQ1-024

Section: MCQ

Type: mcq

Unit: Unit 1, Exploring One-Variable Data and Collecting Data

Skill: Analyze Data

Prompt:

Scores on a spatial-memory task have mean 68 points and standard deviation 7 points. A participant scored 82 points. What is the participant's z -score, and what does it mean?

- A) $z = 1$; the score is 1 standard deviation above the mean.
- B) $z = 2$; the score is 2 points above the mean.
- C) $z = 2$; the score is 2 standard deviations above the mean.
- D) $z = 14$; the score is 14 standard deviations above the mean.

APSTAT-MCQ1-025

Section: MCQ

Type: mcq

Unit: Unit 4, Inference for Quantitative Data: Means

Skill: Collect Data

Prompt:

A sports scientist wants to compare two shoe insoles using the time each runner can maintain a fixed treadmill pace. Which design most appropriately creates paired data?

- A) Each runner tests both insoles in a randomly assigned order, and the two times for each runner are compared.
- B) Runners choose which insole they prefer, and mean times are compared between the two preference groups.
- C) A random half of the runners test one insole, and the other half test the second insole.
- D) All runners test the first insole on Monday and the second insole on Friday.

APSTAT-MCQ1-026

Section: MCQ

Type: mcq

Unit: Unit 2, Probability, Random Variables, and Probability Distributions

Skill: Analyze Data

Prompt:

A quality check independently flags each circuit board with probability 0.20. If 10 boards are checked, what is the probability that exactly 2 are flagged?

- A) 0.107
- B) 0.201
- C) 0.268
- D) 0.302

APSTAT-MCQ1-027

Section: MCQ

Type: mcq

Unit: Unit 1, Exploring One-Variable Data and Collecting Data

Skill: Collect Data

Prompt:

A transit researcher wants a systematic sample of riders leaving a station during one day. The researcher expects about 2,000 riders and wants a sample of 100. Which procedure correctly implements the sample?

- A) Survey the first 100 riders after the station opens.
- B) Choose every 20th rider beginning with rider number 20.
- C) Randomly choose a starting position from 1 through 20, then survey that rider and every 20th rider afterward.
- D) Divide riders into 20 time periods and survey the five most accessible riders in each period.

APSTAT-MCQ1-028

Section: MCQ

Type: mcq

Unit: Unit 3, Inference for Categorical Data: Proportions

Skill: Interpret Results

Prompt:

A randomized experiment compares the proportions of shoppers who choose a product under a redesigned label and under the current label. A two-sided test gives p-value 0.032. At $\alpha = 0.05$, which conclusion is appropriate?

- A) There is a 3.2% chance that the redesigned label has no effect.
- B) Reject H_0 ; the data provide statistically significant evidence that the choice proportions differ between the two labels.
- C) Fail to reject H_0 because 0.032 is greater than 0.025 for a two-sided test.
- D) Reject H_0 and conclude that the redesigned label increases the choice proportion.

APSTAT-MCQ1-029

Section: MCQ

Type: mcq

Unit: Unit 1, Exploring One-Variable Data and Collecting Data

Skill: Formulate Questions

Prompt:

A city will inspect a probability sample of city-owned trees to study pruning needs. Which parameter should be defined to answer the question, "What proportion of all city-owned trees currently need pruning?"

- A) The number of sampled trees that need pruning
- B) The proportion of sampled trees that need pruning
- C) The mean number of branches removed from sampled trees
- D) The proportion p of all city-owned trees that currently need pruning

APSTAT-MCQ1-030

Section: MCQ

Type: mcq

Unit: Unit 4, Inference for Quantitative Data: Means

Skill: Analyze Data

Prompt:

Six technicians each complete the same repair using an old checklist and a new checklist. For each technician, the difference old time – new time, in minutes, is:

2, -1, 4, 3, 2, 5

For a paired t-test of $H_0: \mu_d = 0$, what is the approximate test statistic?

- A) $t = 2.95$
- B) $t = 1.21$
- C) $t = 5.92$
- D) $t = 7.24$

APSTAT-MCQ1-031

Section: MCQ

Type: mcq

Unit: Unit 2, Probability, Random Variables, and Probability Distributions

Skill: Analyze Data

Prompt:

Delivery times for a local courier are approximately normal with mean 32 minutes and standard deviation 4 minutes. What proportion of deliveries take between 28 and 38 minutes? Use $\Phi(-1.00) = 0.1587$ and $\Phi(1.50) = 0.9332$.

- A) 0.0919
- B) 0.6827
- C) 0.7745
- D) 0.8413

APSTAT-MCQ1-032

Section: MCQ

Type: mcq

Unit: Unit 3, Inference for Categorical Data: Proportions

Skill: Analyze Data

Prompt:

A researcher wants a 95% confidence interval for an unknown population proportion with margin of error at most 0.02. With no prior estimate of the proportion, what minimum sample size should be used? Use $z^* = 1.96$.

- A) 601
- B) 2401
- C) 2450
- D) 2500

APSTAT-MCQ1-033

Section: MCQ

Type: mcq

Unit: Unit 1, Exploring One-Variable Data and Collecting Data

Skill: Collect Data

Prompt:

A town posts a QR code on street signs asking residents, "How disruptive is nighttime construction noise?" Participation is optional. What is the greatest threat to using the responses to estimate the opinions of all town residents?

- A) Voluntary-response bias, because residents with strong reactions may be more likely to participate
- B) Undercoverage, because every resident has the same chance to see a street sign
- C) Response bias, because the question requires a numerical answer
- D) Nonresponse bias, because a random sample of selected residents might decline to answer

APSTAT-MCQ1-034

Section: MCQ

Type: mcq

Unit: Unit 5, Regression Analysis

Skill: Analyze Data

Prompt:

Questions APSTAT-MCQ1-034 through APSTAT-MCQ1-036 use the following stimulus. For 18 batches of coffee beans, x is roasting time in minutes and y is an acidity score. Computer output gives the least-squares line $\hat{y} = 92.4 - 3.10x$, with $r = -0.86$ and $r^2 = 0.740$. One batch roasted for 12 minutes had an observed acidity score of 57.0.

What acidity score does the regression line predict for a batch roasted for 12 minutes?

- A) 37.2
- B) 57.0
- C) 55.2
- D) 89.3

APSTAT-MCQ1-035

Section: MCQ

Type: mcq

Unit: Unit 5, Regression Analysis

Skill: Analyze Data

Prompt:

Questions APSTAT-MCQ1-034 through APSTAT-MCQ1-036 use the following stimulus. For 18 batches of coffee beans, x is roasting time in minutes and y is an acidity score. Computer output gives the least-squares line $\hat{y} = 92.4 - 3.10x$, with $r = -0.86$ and $r^2 = 0.740$. One batch roasted for 12 minutes had an observed acidity score of 57.0.

What is the residual for the batch roasted for 12 minutes?

- A) -35.4
- B) -1.8
- C) 1.5
- D) 1.8

APSTAT-MCQ1-036

Section: MCQ

Type: mcq

Unit: Unit 5, Regression Analysis

Skill: Interpret Results

Prompt:

Questions APSTAT-MCQ1-034 through APSTAT-MCQ1-036 use the following stimulus. For 18 batches of coffee beans, x is roasting time in minutes and y is an acidity score. Computer output gives the least-squares line $\hat{y} = 92.4 - 3.10x$, with $r = -0.86$ and $r^2 = 0.740$. One batch roasted for 12 minutes had an observed acidity score of 57.0.

Which statement best describes the relationship summarized by the output?

- A) Roasting time explains 86% of the variation in acidity score, and the relationship is positive.
- B) There is a strong negative linear association, and 74.0% of the variation in acidity score is explained by its linear relationship with roasting time.
- C) Each additional minute of roasting causes acidity to decrease by exactly 3.10 points.
- D) Seventy-four percent of batches have acidity scores below their predicted values.

APSTAT-MCQ1-037

Section: MCQ

Type: mcq

Unit: Unit 4, Inference for Quantitative Data: Means

Skill: Formulate Questions

Prompt:

A warehouse records each worker's packing time before and after a new workstation layout. Define $d = \text{after} - \text{before}$. The manager wants to determine whether the layout reduces the population mean packing time. Which hypotheses are appropriate?

- A) $H_0: \mu_d < 0$; $H_a: \mu_d = 0$
- B) $H_0: \mu_d = 0$; $H_a: \mu_d > 0$
- C) $H_0: \mu_d = 0$; $H_a: \mu_d < 0$
- D) $H_0: \mu_{\text{after}} = \mu_{\text{before}}$; $H_a: \mu_{\text{after}} \neq \mu_{\text{before}}$, treating the samples as independent

APSTAT-MCQ1-038

Section: MCQ

Type: mcq

Unit: Unit 3, Inference for Categorical Data: Proportions

Skill: Analyze Data

Prompt:

A distributor tests $H_0: p = 0.04$ against $H_a: p > 0.04$, where p is the proportion of all incoming shipments containing a damaged item. Which situation is a Type I error?

- A) Concluding that more than 4% of shipments contain a damaged item when the true proportion is 4%
- B) Concluding that the proportion is 4% when the true proportion is greater than 4%
- C) Failing to conclude that the proportion exceeds 4% when the true proportion is 4%
- D) Concluding that fewer than 4% of shipments contain a damaged item when the true proportion exceeds 4%

APSTAT-MCQ1-039

Section: MCQ

Type: mcq

Unit: Unit 1, Exploring One-Variable Data and Collecting Data

Skill: Analyze Data

Prompt:

Annual donations to a small arts nonprofit have a strongly right-skewed distribution because a few donors make very large gifts. Which pair of statistics is generally most appropriate for describing a typical donation and its spread?

- A) Mean and range
- B) Mean and standard deviation
- C) Median and standard deviation
- D) Median and IQR

APSTAT-MCQ1-040

Section: MCQ

Type: mcq

Unit: Unit 2, Probability, Random Variables, and Probability Distributions

Skill: Interpret Results

Prompt:

The fill weight of individual cereal boxes has population mean $\mu = 500$ grams and population standard deviation $\sigma = 12$ grams. The population distribution is moderately right-skewed. For random samples of $n = 36$ boxes, which statement best describes the sampling distribution of $\bar{x}$?

- A) It is right-skewed with mean 500 grams and standard deviation 12 grams.
- B) It is approximately normal with mean 500 grams and standard deviation 2 grams.

- C) It is exactly normal with mean 36 grams and standard deviation 2 grams.
- D) Its shape cannot be approximated because the population is not normal.

APSTAT-MCQ1-041

Section: MCQ

Type: mcq

Unit: Unit 4, Inference for Quantitative Data: Means

Skill: Interpret Results

Prompt:

A biologist repeatedly takes random samples of the same size from a fish population and constructs a 90% t-confidence interval for the population mean length from each sample. What does the 90% confidence level mean?

- A) For every sample, 90% of sampled fish lengths lie inside that sample's interval.
- B) There is a 90% chance that the population mean changes to fall inside a computed interval.
- C) In repeated sampling, about 90% of intervals produced by this method will contain the population mean length.
- D) Ninety percent of all fish have lengths between the endpoints of any one interval.

APSTAT-MCQ1-042

Section: MCQ

Type: mcq

Unit: Unit 3, Inference for Categorical Data: Proportions

Skill: Formulate Questions

Prompt:

Last year, 48% of a transit system's riders used mobile tickets. The agency takes a random sample this year to investigate whether the current population proportion p differs from last year's value. Which alternative hypothesis represents the research question?

- A) $H_a: p \neq 0.48$
- B) $H_a: p = 0.48$
- C) $H_a: p > 0.48$
- D) $H_a: p < 0.48$

PART B, ANSWER KEY AND EXPLANATIONS

APSTAT-MCQ1-001, Correct: B, Correct Answer: The value 515 is a potential outlier because it is greater than the upper fence of 470 cycles., Explanation: $Q1 = 420$ and $Q3 = 440$, so $IQR = 20$. The upper fence is $440 + 1.5(20) = 470$, and $515 > 470$. The lower fence is 390, so 410 is not flagged.

APSTAT-MCQ1-002, Correct: B, Correct Answer: The data provide evidence that the premium-plan population renewal proportion is between 4 and 18 percentage points higher than the standard-plan proportion., Explanation: The entire interval for $p_P - p_S$ is positive. The interval estimates a difference

between population proportions, not individual outcomes, monthly differences, or the already observed sample difference.

APSTAT-MCQ1-003, Correct: D, Correct Answer: 0.600, Explanation: Conditioning on a torn label restricts the denominator to the 30 torn-label parcels. Thus $P(\text{late scan} \mid \text{torn label}) = 18 \div 30 = 0.600$.

APSTAT-MCQ1-004, Correct: C, Correct Answer: Take separate simple random samples of morning and evening riders, with sample sizes proportional to their shares of weekday ridership., Explanation: Stratifying by time of day guarantees representation of both groups, and proportional allocation lets the combined sample represent overall weekday ridership. The other methods use convenience, a small number of clusters, or voluntary response.

APSTAT-MCQ1-005, Correct: A, Correct Answer: $t = 2.29$, Explanation: The one-sample statistic is $t = (\bar{x} - \mu_0) \div (s/\sqrt{n}) = (18.6 - 17) \div (3.5/\sqrt{25}) = 1.6 \div 0.7 \approx 2.29$.

APSTAT-MCQ1-006, Correct: A, Correct Answer: -4.2 minutes, Explanation: A residual is observed $-$ predicted, so $37.0 - 41.2 = -4.2$ minutes. The negative residual means unloading took 4.2 minutes less than predicted.

APSTAT-MCQ1-007, Correct: D, Correct Answer: $H_0: p = 0.70$; $H_a: p > 0.70$, Explanation: The equality belongs in H_0 , and "more than 70%" makes the alternative right-tailed.

APSTAT-MCQ1-008, Correct: C, Correct Answer: A cluster sample using buildings as clusters, Explanation: Whole groups are selected at random and every apartment in each selected group is inspected. That is cluster sampling; a stratified design would sample some apartments from every building or stratum.

APSTAT-MCQ1-009, Correct: B, Correct Answer: 0.75 failure, Explanation: $E(X) = \sum xP(X = x) = 0(0.50) + 1(0.30) + 2(0.15) + 3(0.05) = 0.75$ failure per week.

APSTAT-MCQ1-010, Correct: D, Correct Answer: 0.89 failure, Explanation: $E(X^2) = 0^2(0.50) + 1^2(0.30) + 2^2(0.15) + 3^2(0.05) = 1.35$. Therefore $\sigma_X = \sqrt{1.35 - (0.75)^2} = \sqrt{0.7875} \approx 0.89$ failure.

APSTAT-MCQ1-011, Correct: A, Correct Answer: Over many weeks, the average number of pump failures per week will approach 0.75., Explanation: An expected value is a long-run average over many repetitions. It need not be an outcome possible in a single week and does not fix the count in any short period.

APSTAT-MCQ1-012, Correct: C, Correct Answer: The data indicate that the population mean weekly commuting time after the change is 0.4 to 1.8 hours lower than before., Explanation: Because $d = \text{after} - \text{before}$ and the interval is $(-1.8, -0.4)$, the population mean after-time is estimated to be 0.4 to 1.8 hours lower. The interval concerns a population mean difference, not every student.

APSTAT-MCQ1-013, Correct: A, Correct Answer: The mean increases by 12, and the standard deviation is unchanged., Explanation: Adding the same constant to every value shifts the center by that constant but does not change any distance from the mean, so spread is unchanged.

APSTAT-MCQ1-014, Correct: B, Correct Answer: (0.361, 0.499), Explanation: $\hat{p} = 86/200 = 0.43$. The standard error is $\sqrt{0.43(0.57)/200} \approx 0.0350$, so the 95% interval is $0.43 \pm 1.96(0.0350) \approx 0.43 \pm 0.0687$

= (0.361, 0.499).

APSTAT-MCQ1-015, Correct: D, Correct Answer: Sixty-four percent of the variation in weekly repair cost is explained by its linear relationship with weekly operating hours., Explanation: r^2 is the proportion of variation in the response explained by the least-squares regression on the explanatory variable. It is not a correlation, a percentage of exact predictions, or evidence of causation.

APSTAT-MCQ1-016, Correct: C, Correct Answer: Within each slope band, randomly assign equal numbers of plots to the three treatments., Explanation: Treating slope bands as blocks controls location-related variation, while random assignment within each block supports a causal comparison of treatments.

APSTAT-MCQ1-017, Correct: B, Correct Answer: Mean 0.30 and standard deviation 0.0229, Explanation: The sampling distribution is centered at $p = 0.30$, and $\sigma_{\hat{p}} = \sqrt{[p(1 - p)/n]} = \sqrt{[0.30(0.70)/400]} \approx 0.0229$.

APSTAT-MCQ1-018, Correct: A, Correct Answer: 1.13, Explanation: For independent samples, $SE(\bar{x}_A - \bar{x}_B) = \sqrt{[(4.8^2/36) + (5.6^2/49)]} = \sqrt{(0.64 + 0.64)} = \sqrt{1.28} \approx 1.13$.

APSTAT-MCQ1-019, Correct: D, Correct Answer: A causal effect can be concluded for the participating clinics, but generalization to all clinics is not justified., Explanation: Random assignment permits a causal conclusion for the participating clinics. Because the clinics volunteered rather than being randomly sampled, they may not represent all clinics.

APSTAT-MCQ1-020, Correct: D, Correct Answer: Their medians are equal, and the middle half of Courier A's distances is more variable., Explanation: Both medians are 6.5 km. $IQR_A = 11 - 3 = 8$ km and $IQR_B = 8 - 5 = 3$ km, so Courier A has greater spread in the middle 50%; Courier B's outlier does not determine its IQR.

APSTAT-MCQ1-021, Correct: A, Correct Answer: -2.70 watts per centimeter, Explanation: The least-squares slope is $b_1 = r(s_y/s_x) = -0.72(9.0/2.4) = -2.70$ watts per centimeter.

APSTAT-MCQ1-022, Correct: C, Correct Answer: $P(X \geq 3)$, Explanation: "At least 3" includes 3, 4, 5, 6, 7, and 8, so the desired event is $X \geq 3$ rather than $X = 3$ or $X \leq 3$.

APSTAT-MCQ1-023, Correct: B, Correct Answer: 2.35, Explanation: $\hat{p}_1 = 78/120 = 0.65$, $\hat{p}_2 = 60/120 = 0.50$, and the pooled proportion is $138/240 = 0.575$. Thus $z = (0.65 - 0.50) \div \sqrt{[0.575(0.425)(1/120 + 1/120)]} \approx 0.15/0.0638 \approx 2.35$.

APSTAT-MCQ1-024, Correct: C, Correct Answer: $z = 2$; the score is 2 standard deviations above the mean., Explanation: $z = (82 - 68)/7 = 2$. A z-score measures distance from the mean in standard-deviation units, not raw points.

APSTAT-MCQ1-025, Correct: A, Correct Answer: Each runner tests both insoles in a randomly assigned order, and the two times for each runner are compared., Explanation: Each runner serves as a matched unit, controlling runner-to-runner differences. Randomizing order helps prevent order effects from being confounded with insole type.

APSTAT-MCQ1-026, Correct: D, Correct Answer: 0.302, Explanation: With $X \sim \text{Binomial}(10, 0.20)$, $P(X = 2) = C(10, 2)(0.20)^2(0.80)^8 = 45(0.04)(0.1678) \approx 0.302$.

APSTAT-MCQ1-027, Correct: C, Correct Answer: Randomly choose a starting position from 1 through 20, then survey that rider and every 20th rider afterward., Explanation: The sampling interval is $2,000/100 = 20$. A random start among the first 20 positions is necessary; always starting at rider 20 is not random.

APSTAT-MCQ1-028, Correct: B, Correct Answer: Reject H_0 ; the data provide statistically significant evidence that the choice proportions differ between the two labels., Explanation: Since $0.032 < 0.05$, H_0 is rejected. A two-sided test supports a difference but, without the observed direction, does not by itself justify saying the redesigned label increases the proportion. The p-value is not the probability H_0 is true.

APSTAT-MCQ1-029, Correct: D, Correct Answer: The proportion p of all city-owned trees that currently need pruning, Explanation: A parameter describes the population. Here p is the unknown proportion among all city-owned trees; the corresponding sample proportion would be a statistic used to estimate p .

APSTAT-MCQ1-030, Correct: A, Correct Answer: $t = 2.95$, Explanation: The differences have $\bar{x}_d = 2.5$ and $s_d = \sqrt{(21.5/5)} \approx 2.074$. Therefore $t = (2.5 - 0) \div (2.074/\sqrt{6}) \approx 2.95$.

APSTAT-MCQ1-031, Correct: C, Correct Answer: 0.7745, Explanation: The endpoints correspond to $z = (28 - 32)/4 = -1.00$ and $z = (38 - 32)/4 = 1.50$. The probability is $\Phi(1.50) - \Phi(-1.00) = 0.9332 - 0.1587 = 0.7745$.

APSTAT-MCQ1-032, Correct: B, Correct Answer: 2401, Explanation: With no prior estimate, use $p^* = 0.50$ for the largest required size. Set $1.96\sqrt{[0.25/n]} \leq 0.02$, giving $n \geq 0.25(1.96/0.02)^2 = 2401$. Thus the minimum sample size is 2401.

APSTAT-MCQ1-033, Correct: A, Correct Answer: Voluntary-response bias, because residents with strong reactions may be more likely to participate, Explanation: Respondents decide whether to join, and people most affected by the noise may respond at a higher rate. No random sample was selected, so calling the issue nonresponse from a sampled group would be incorrect.

APSTAT-MCQ1-034, Correct: C, Correct Answer: 55.2, Explanation: Substitute $x = 12$ into the line: $\hat{y} = 92.4 - 3.10(12) = 92.4 - 37.2 = 55.2$ acidity points.

APSTAT-MCQ1-035, Correct: D, Correct Answer: 1.8, Explanation: Residual = observed – predicted = $57.0 - 55.2 = 1.8$ acidity points. A positive residual means the observed score is above the prediction.

APSTAT-MCQ1-036, Correct: B, Correct Answer: There is a strong negative linear association, and 74.0% of the variation in acidity score is explained by its linear relationship with roasting time., Explanation: $r = -0.86$ indicates a strong negative linear association, and $r^2 = 0.740$ gives the explained proportion of response variation. Regression alone does not establish causation.

APSTAT-MCQ1-037, Correct: C, Correct Answer: $H_0: \mu_d = 0$; $H_a: \mu_d < 0$, Explanation: If the layout reduces time, after – before tends to be negative. The null uses equality at zero, and the paired differences—not two independent samples—are analyzed.

APSTAT-MCQ1-038, Correct: A, Correct Answer: Concluding that more than 4% of shipments contain a damaged item when the true proportion is 4%, Explanation: A Type I error rejects a true null hypothesis. Here that means concluding $p > 0.04$ when $H_0: p = 0.04$ is true.

APSTAT-MCQ1-039, Correct: D, Correct Answer: Median and IQR, Explanation: The median and IQR are resistant to extreme high donations and are therefore better summaries of center and spread for a strongly right-skewed distribution.

APSTAT-MCQ1-040, Correct: B, Correct Answer: It is approximately normal with mean 500 grams and standard deviation 2 grams., Explanation: For $n = 36$, the central limit theorem supports an approximately normal sampling distribution despite moderate population skew. Its mean is $\mu_{\bar{x}} = 500$ and its standard deviation is $\sigma_{\bar{x}} = 12/\sqrt{36} = 2$ grams.

APSTAT-MCQ1-041, Correct: C, Correct Answer: In repeated sampling, about 90% of intervals produced by this method will contain the population mean length., Explanation: Confidence level describes the long-run success rate of the interval-producing method. The fixed population mean does not change from sample to sample.

APSTAT-MCQ1-042, Correct: A, Correct Answer: $H_a: p \neq 0.48$, Explanation: The word “differs” allows either an increase or a decrease, so the alternative is two-sided. Equality belongs in H_0 , not H_a .

SERIES DO-NOT-REPEAT BANK UPDATE, APSTAT-MCQ1

· APSTAT-MCQ1-001 to 005: Unit 1 battery-cycle raw data use quartiles, IQR fences, and a high outlier; Unit 3 independent subscriber-plan renewal proportions use a fully positive two-proportion confidence interval and distinguish population differences from individual outcomes; Unit 2 parcel torn-label-by-late-scan count table uses conditional probability with the restricted denominator; Unit 1 ferry satisfaction uses proportionally allocated strata for morning and evening riders and targets convenience, cluster, and voluntary-response misconceptions; Unit 4 museum gallery times use a one-sample t statistic from n , $\bar{x}$, s , and μ_0 . Misconceptions include using standard deviations for the IQR rule, treating confidence as probability after computation, using the grand total for conditional probability, and omitting $\sqrt{n}$ from standard error.

· APSTAT-MCQ1-006 to 010: Unit 5 truck-unloading regression uses observed minus predicted to obtain a negative residual; Unit 3 self-checkout success threshold formulates a right-tailed one-proportion test; Unit 1 apartment smoke-detector inspection selects entire buildings as clusters; Unit 2 three-question greenhouse pump-failure distribution begins with expected value and then variance from $E(X^2) - [E(X)]^2$. Misconceptions include reversing residual sign, putting inequality in H_0 , confusing clusters with strata, and treating expected value as the most likely outcome.

· APSTAT-MCQ1-011 to 015: Unit 2 greenhouse pump failures interpret expected value as a long-run weekly average; Unit 4 paired student commuting differences after – before interpret an entirely negative t -interval; Unit 1 lake-temperature additive calibration distinguishes shift in center from unchanged spread; Unit 3 reusable-utensil orders construct a one-proportion z -interval; Unit 5 machine operating hours and repair cost interpret r^2 as explained response variation. Misconceptions include expecting an exact short-run count, applying a mean interval to individuals, changing standard deviation after adding a constant, omitting z^* , and interpreting r^2 as causation or correlation.

· APSTAT-MCQ1-016 to 020: Unit 1 hillside soil-treatment experiment blocks by slope band with random assignment within blocks; Unit 2 promised-day delivery $\hat{p}$ sampling distribution uses center p and standard deviation $\sqrt{p(1-p)/n}$; Unit 4 independent samples A and B compute the standard error of $\bar{x}_A - \bar{x}_B$; Unit 3 volunteer clinics randomly assigned to reminder conditions distinguish causal scope from population generalization; Unit 1 courier side-by-side boxplot summaries compare equal

medians and IQRs despite an outlier. Misconceptions include confounding block with treatment, reporting count rather than proportion parameters, adding standard errors, equating random assignment with random sampling, and letting an outlier determine IQR.

- APSTAT-MCQ1-021 to 025: Unit 5 insulation thickness and heat loss derive least-squares slope from $r(s_y/s_x)$; Unit 2 independent customer returns formulate the “at least 3” binomial event; Unit 3 district transit support calculates a pooled two-proportion z statistic; Unit 1 spatial-memory score converts raw distance to a z-score and interprets standard-deviation units; Unit 4 runner insole comparison uses randomized treatment order in a paired design. Misconceptions include inverting s_y/s_x , confusing $\geq$ with $=$ or $\leq$, failing to pool under H_0 , treating a z-score as raw points, and calling two independent groups paired.

- APSTAT-MCQ1-026 to 030: Unit 2 circuit-board flags calculate an exact binomial probability with the combination factor; Unit 1 station riders use a random start and every 20th rider for a systematic sample; Unit 3 randomized product-label experiment interprets a two-sided p-value at $\alpha = 0.05$ without treating it as $P(H_0)$; Unit 1 city tree pruning identifies the population proportion parameter rather than a sample statistic; Unit 4 paired technician repair-time differences calculate $\bar{x}_d$, s_d , standard error, and a t statistic. Misconceptions include omitting the binomial coefficient, using a fixed nonrandom systematic start, dividing α for the final p-value comparison, confusing statistic with parameter, and treating paired observations as independent.

- APSTAT-MCQ1-031 to 035: Unit 2 normally distributed courier delivery times use two unequal z-boundaries and subtract cumulative probabilities; Unit 3 conservative one-proportion margin-of-error planning uses $p^* = 0.50$ and rounds sample size upward; Unit 1 optional street-sign QR survey about construction noise identifies voluntary-response bias; Unit 5 coffee-roasting stimulus uses the line $\hat{y} = 92.4 - 3.10x$ to predict acidity at 12 minutes and compute observed-minus-predicted residual. Misconceptions include adding tail areas, mishandling the 0.25 factor in sample-size planning, labeling a nonrandom volunteer poll as nonresponse, and reversing residual sign.

- APSTAT-MCQ1-036 to 040: Unit 5 coffee-roasting output interprets $r = -0.86$ and $r^2 = 0.740$ without causal language; Unit 4 warehouse packing times formulate a left-tailed paired-mean test for $d = \text{after} - \text{before}$; Unit 3 damaged-shipment rate test identifies a Type I error under $H_0: p = 0.04$; Unit 1 right-skewed nonprofit donations select resistant median and IQR; Unit 2 cereal-box sample means apply the central limit theorem with $\mu_{\bar{x}} = 500$ and $\sigma_{\bar{x}} = 2$. Misconceptions include reading r as explained percentage, ignoring the defined difference direction, confusing Type I and Type II errors, pairing median with standard deviation, and using σ rather than $\sigma/\sqrt{n}$.

- APSTAT-MCQ1-041 to 042: Unit 4 repeated fish-length samples interpret 90% confidence as the long-run capture rate of the interval method; Unit 3 current mobile-ticket use versus last year's 0.48 formulates a two-sided alternative from “differs.” Misconceptions include assigning probability to a fixed parameter after an interval is computed, treating a mean interval as covering individual fish, and choosing a one-sided alternative without directional wording.

- Unit counts for APSTAT-MCQ1: Unit 1 = 11; Unit 2 = 9; Unit 3 = 9; Unit 4 = 7; Unit 5 = 6. Skill counts: Formulate Questions = 5; Collect Data = 7; Analyze Data = 21; Interpret Results = 9. Stimulus sets: greenhouse pump-failure distribution, APSTAT-MCQ1-009 to 011; coffee-roasting regression output, APSTAT-MCQ1-034 to 036.